

Lec 34

Convergence of R.V.s.

Recap

Convergence of sequence of real numbers $(a_n)_{n \geq 1}$

$a_n \rightarrow a$ if $\forall \epsilon > 0, \exists N \in \mathbb{N}$ st

is

$$|a_n - a| \leq \epsilon \quad \forall n \geq N$$

$n \geq N(\epsilon)$

$a_n \not\rightarrow a$

$a_1, a_2, a_3, \dots$

$a_n \rightarrow a$
 $n \geq N(\epsilon)$

$\exists \epsilon > 0, \forall N \in \mathbb{N}, \exists n \geq N$

$$|a_n - a| > \epsilon$$

Convergence of series.

$$S_n = \sum_{i=1}^n a_i$$

$$S_n \rightarrow s \quad [\text{series } S_n \text{ converges to } s]$$

$$\text{Let } \hat{S}_n = \sum_{i=n}^{\infty} a_i \quad \text{and } S_n = \sum_{i=1}^n a_i$$

$$\textcircled{1} \quad \text{If } S_n \rightarrow s, \quad \hat{S}_n \rightarrow 0$$

$$\textcircled{2} \quad \text{If } S_n \rightarrow s, \quad a_n \rightarrow 0$$

$$S = \sum_{i=1}^{\infty} a_i, \quad \hat{S}_n = S - S_{n-1}$$

$$\lim_{n \rightarrow \infty} \hat{S}_n = S - \lim_{n \rightarrow \infty} S_{n-1} = S - S = 0$$

$$a_n = S_n - S_{n-1}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} S_{n-1} \\ &= S - S = 0 \end{aligned}$$

Pointwise convergence

$X_1, X_2, X_3, \dots$ are sequence of
r.v.s defined over $(\Omega, \mathcal{F}, P)$.

' X_n 's are said to converge
pointwise to a R.V. X if

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \quad \omega \in \Omega$$

Almost-Sure Convergence

$$\lim_{n \rightarrow \infty} S_n = 1$$

$$S = 0$$

rgence

are sequence of
($\Omega, \mathcal{F}, P$).

to converge

$\forall X$ if

$$= X(\omega) \quad \omega \in \Omega$$

$$P(\{\omega: \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1$$

then $X_n \xrightarrow{\text{almost sure (a.s.)}} X$

a.

Examples: $\Omega = [0, 1]$

$$X_n(\omega) = \begin{cases} \omega^n & \omega \in [0, 1] \end{cases}$$

$$X_n: \Omega \rightarrow \mathbb{R}$$
$$X(\omega) = \begin{cases} 0 & \omega \in [0, 1] \\ 1 & \text{otherwise} \end{cases}$$

$X(\omega)$



$$\lim_{n \rightarrow \infty}$$

$\nRightarrow X$

gence

$$= X(\omega)\} = 1$$

X

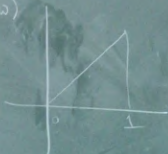
$$[0, 1]$$

$$\omega \in [0, 1]$$

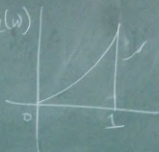
$$\omega \in [0, 1]$$

otherwise

$$X(\omega)$$



$$X_2(\omega)$$



$$\lim_{n \rightarrow \infty} X_n(\omega) = \lim_{n \rightarrow \infty} \omega^n$$

$$= \begin{cases} 0 & \omega \in [0, 1) \\ 1 & \text{otherwise} \end{cases}$$

$$= X(\omega)$$

$$\Rightarrow X_n \rightarrow X \text{ pointwise}$$

$$\textcircled{b} \hat{X}(\omega) =$$

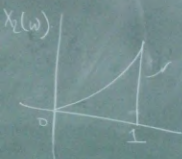
$$X_n \xrightarrow{d.s.}$$

$$\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}$$

$$P([0, 1])$$

$$\Rightarrow P(\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\})$$

$$\Rightarrow Y$$



$$\lim_{n \rightarrow \infty} \omega^n$$

$$0 \quad \omega \in [0, 1)$$

$$1 \quad \text{otherwise}$$

$$X(\omega)$$

$$X \quad \text{pointwise}$$

$$\textcircled{b} \quad \hat{X}(\omega) = 0 \quad \omega \in [0, 1]$$

$$X_n \xrightarrow{a.s.} \hat{X}$$

$$\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = \hat{X}(\omega)\} = [0, 1]$$

$$P([0, 1]) = 1$$

$$\Rightarrow P(\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = \hat{X}(\omega)\}) = 1$$

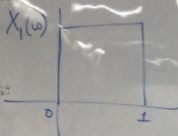
$$\Rightarrow X_n \xrightarrow{a.s.} \hat{X}$$

$$X_n \rightarrow \hat{X} \quad \text{pointwise}$$

Lec 34

Convergence of R.V.s

② Moving-Shrinking rectangles example.



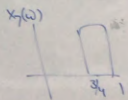
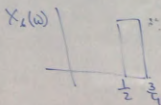
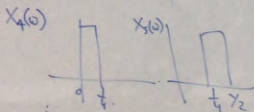
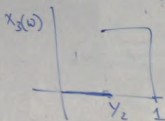
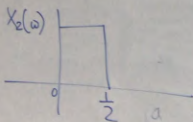
$$(X_1(\omega) \ X_2(\omega) \ X_3(\omega) \ \dots)$$

$$= (1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ \dots)$$

"1 will appear infinitely often"

This is indeed true for every ω .

$X_1(\omega) \ X_2(\omega) \ X_3(\omega) \ \dots$
doesn't converge.



If $X_n \rightarrow X$ almost surely then

$$P(\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1.$$

$$A = \{\omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega)\}$$

Bad set

If $\omega \in A$, $X_n(\omega) \not\rightarrow X(\omega)$.

$\exists \epsilon > 0$, $\forall N \in \mathbb{N}$, $\exists n \geq N$ st

$$|X_n(\omega) - X(\omega)| > \epsilon.$$



$|X_n(\omega) - X(\omega)| > \epsilon$ infinitely often (i.o.)

$$n_1, n_2, \dots, |X_{n_i}(\omega) - X(\omega)| > \epsilon.$$

$$A_n = \left\{ \omega \in \Omega \mid |X_n(\omega) - X(\omega)| > \varepsilon \right\}$$

$$\limsup A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n$$

$$\omega \in \limsup A_n$$

$$\omega \in \bigcup_{n=1}^{\infty} A_n \text{ \& \& } \omega \in \bigcup_{n=2}^{\infty} A_n \text{ \& \& \& \& \& }$$

Let n_1 be st

$$\omega \in A_{n_1}, \quad \omega \in A_{n_2}, \quad \dots$$

To show almost sure convergence it is enough to show:

$$\text{that } P(\limsup A_n) = 0$$



$$A = \left\{ \omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega) \right\}$$

$$A^c = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} \left\{ \omega : |X_n(\omega) - X(\omega)| < \frac{1}{n} \right\}$$

For any ϵ if $P(A^c) = 0$ then
 $P(A) = 0$

to show

$$\textcircled{b} \bigwedge \omega$$

$$X_n$$

$$\left\{ \omega : \right.$$

$$P($$

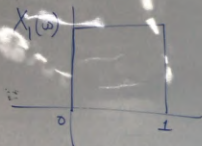
$$\Rightarrow P($$

$$=$$

Lec 34

Convergence of R.V.s.

② Moving-Shrinking rectangles example.



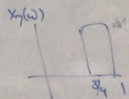
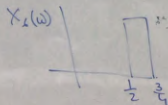
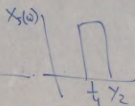
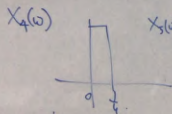
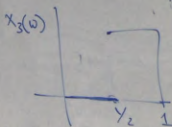
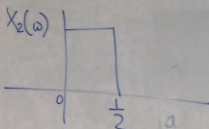
$$(X_1(\omega) \ X_2(\omega) \ X_3(\omega) \ \dots)$$

$$= (1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ \dots \ 0 \ 1 \ \dots)$$

"1 will appear infinitely often"

This is indeed true for every ω .

$X_1(\omega) \ X_2(\omega) \ X_3(\omega) \ \dots$
doesn't converge.



If $X_n \rightarrow X$ almost surely then

$$P\left(\left\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

$$A = \left\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega)\right\}$$

Bad set

If $\omega \in A$, $X_n(\omega) \not\rightarrow X(\omega)$.

$$\boxed{\exists \epsilon > 0, \forall N \in \mathbb{N}, \exists n \geq N \text{ st } |X_n(\omega) - X(\omega)| > \epsilon.}$$



$|X_n(\omega) - X(\omega)| > \epsilon$ infinitely often (i.o.)

$$n_1, n_2, \dots, |X_{n_i}(\omega) - X(\omega)| > \epsilon$$

$$A = \left\{ \omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega) \right\}$$

$$A_\epsilon = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} \left\{ \omega : |X_n(\omega) - X(\omega)| \geq \epsilon \right\}$$

For any ϵ if $P(A_\epsilon) = 0$ then

$$P(A) = 0$$

Borel-Cantelli Lemmas

gh to show:

Lemma: Let $A_1, A_2, \dots$ be a sequence of events st $\sum_{n=1}^{\infty} P(A_n) < \infty$
then $P(\limsup A_n) = 0$

Example

$X_1, X_2, \dots$ are iid Uniform $[0, 1]$ RVs.

$$Y_n = \min(X_1, X_2, \dots, X_n)$$

$$Y_n \xrightarrow{a.s.} \boxed{0}$$

$$\begin{aligned} A_n &= \{\omega : |Y_n(\omega) - 0| > \varepsilon\} \\ &= \{\omega : Y_n(\omega) > \varepsilon\} \end{aligned}$$

$$P(A_n) = P(Y_n > \varepsilon)$$

$$= P(\min\{X_1, \dots, X_n\} > \varepsilon)$$

$$= \prod_{i=1}^n P(X_i > \varepsilon) = (1 - \varepsilon)^n$$

$$\sum_{n=1}^{\infty} P(A_n) < \infty$$

$$\Downarrow$$
$$P(\limsup A_n) = 0$$

$$0 \leq P\left(\bigcap_{N=1}^{\infty} \underbrace{\bigcup_{n \geq N} A_n}_{B_N}\right) = P\left(\bigcap_{N=1}^{\infty} B_N\right) = \lim_{N \rightarrow \infty} P(B_N) \quad \left\{ \begin{array}{l} \text{Continuity} \\ \text{property} \\ \text{corollaries} \end{array} \right.$$

$$B_1 = \bigcup_{n=1}^{\infty} A_n$$

$$B_2 = \bigcup_{n=2}^{\infty} A_n$$

⋮

$$= \lim_{N \rightarrow \infty} P\left(\bigcup_{n \geq N} A_n\right)$$

$$\leq \lim_{N \rightarrow \infty} \sum_{n=N}^{\infty} P(A_n) \quad \left\{ \begin{array}{l} \text{Union} \\ \text{bound} \end{array} \right.$$

$$B_1 \supseteq B_2 \supseteq \dots$$

$$S_n = \sum_{i=1}^n P(A_i)$$

$$S_n = \sum_{i=n}^{\infty} P(A_i)$$

$$= 0$$

$$\lim_{n \rightarrow \infty} S_n = 0$$

$$\text{if } S_n \rightarrow s < \infty$$

$$\Rightarrow P(\limsup A_n) = 0$$

Convergence in probability

$X_1, X_2, \dots$ are said to converge to X in probability sense if

For any $\epsilon > 0$

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$$

Example: Moving-Shrinking rectangles

$$X_n \xrightarrow[\text{probability sense}]{\quad} 0$$

$$a_n = P(X_n > \epsilon)$$

$$a_1 = 1$$

$$a_2 = \frac{1}{2}$$

$$a_3 = \frac{1}{2}$$

$$a_4 = \frac{1}{2^2}$$

$$a_5 = \frac{1}{2^2}$$

$$\lim_{n \rightarrow \infty} a_n = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} P(X_n > \epsilon) = 0$$